

A Chemical Kinetic Study on the Influence of Micro-Mix of Hydrogen (H_2) and Hydrogen Peroxide (H_2O_2) in Ammonia Mixtures.

Report for

B.Tech Project 1

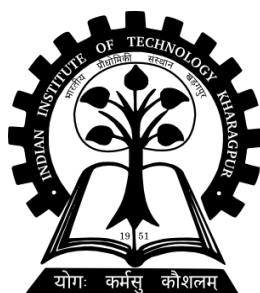
by

Sarthak Fulzele

21ME31039

Under the guidance of

Prof. Vanteru Mahendra Reddy



**DEPARTMENT OF MECHANICAL ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR**

Table of Contents

1. **Nomenclature**
 2. **Chapter 1: Introduction**
 - 2.1. Background
 - 2.2. The Role of Ammonia as a Sustainable Fuel
 - 2.3. Motivation for Research
 - 2.4. Objectives of the Study
 3. **Chapter 2: Literature Review**
 - 3.1. Ammonia Combustion and Emission Characteristics
 - 3.2. Effect of Oxidizer ratio
 - 3.3. Effect of Preheating Temperature
 - 3.4. Combined Effects of Oxidizer Ratio and Preheating Temperature
 - 3.5. Perfectly Stirred Reactor (PSR) Modeling
 4. **Chapter 3: Methodology**
 - 4.1. Simulation Setup
 - 4.2. Heatmap Generation and Plotting Techniques
 - 4.3. Perfectly Stirred Reactor (PSR) Analysis
 5. **Chapter 4: Results**
 - 5.1. Heatmaps of Key Parameters
 - 5.2. Heatmap Generation Methodology
 6. **Chapter 5: Conclusions**
 7. **Chapter 6: Future Prospects**
 - 7.1. Optimization of Additive Concentrations
 - 7.2. Exploration of Alternative Fuels
 - 7.3. Numerical Modelling and Machine Learning Implementation
 - 7.4. Experimental Validation and Scale-Up Studies
 8. **References/Bibliography**
-

Nomenclature

- **NO_x**: Nitrogen Oxides
 - **NH₃**: Ammonia
 - **PSR**: Perfectly Stirred Reactor
-

Abstract

The world is focused on transitioning to cleaner energy to reduce rising emissions from combustion applications. In this context, ammonia stands out as a great alternative due to its high hydrogen content and ease in storage in transportation. Although ammonia has some challenges, like lower reactivity and nitrogen oxides (NO_x) emissions during combustion, researchers are finding the novel technologies. Recent studies show that pure ammonia flames are hard to adopt in traditional combustion systems. However, using fuel blending can improve ammonia combustion compared to methods like preheating and plasma-assisted combustion. Our previous work found that adding hydrogen helps stabilize flames and reduces emissions as compared with the hydrocarbon fuel blending. This study addresses a lack of understanding about the stability of pure ammonia flames under different preheating temperatures and oxygen levels. In this present study, a wide range of numerical simulations were performed to study pure ammonia and with micro-mix of hydrogen (5–15% vol) and hydrogen peroxide (H₂O₂). The range of preheating temperatures of 500–1500 K and oxygen concentrations range of 0.25–21% vol were considered. Around 30,000 simulations were performed with Ansys Chemkin Pro. The addition of 5% hydrogen significantly enhanced flame stability, while further increases to 10% and 15% hydrogen did not yield additional stability but did elevate NO_x emissions. In contrast, adding 5% H₂O₂ notably strengthened flame stability compared to the hydrogen-only cases. To visualize these outcomes, we developed heat maps and emission maps that illustrate the impact of H₂ and H₂O₂ in pure ammonia mixtures. This research not only deepens our understanding of ammonia combustion dynamics but also paves the way for more effective strategies in the pursuit of cleaner, more sustainable energy solutions.

Furthermore, an artificial neural network (ANN) model was developed to predict the outcomes based on the extensive dataset obtained from the simulations. The ANN demonstrated high predictive accuracy, effectively capturing the complex relationships between preheating temperatures, oxygen concentrations, and the influence of hydrogen and H₂O₂ on flame

stability and emissions. This model provides a robust tool for optimizing combustion conditions, offering insights that could further accelerate the adoption of ammonia as a viable alternative fuel in sustainable energy applications.

Chapter 1: Introduction

1.1 Background

The quest for sustainable and cleaner energy sources has driven extensive research into alternative fuels and chemical mixtures. Among these, ammonia (NH_3) has gained attention as a promising candidate due to its carbon-free combustion properties and high hydrogen content (approximately 17.6% by mass). Additionally, ammonia can be easily stored and transported using existing infrastructure developed for agricultural fertilizers (Zhao et al., 2023). However, the conventional challenges associated with ammonia combustion—namely its high ignition temperature and low flame speed—necessitate innovative approaches to enhance its burn efficiency and stability (Kobayashi et al., 2020).

One of the intriguing methods to overcome these limitations is the micro-mixing of hydrogen (H_2) with ammonia. Hydrogen, being a highly reactive fuel with a lower ignition temperature, can significantly enhance the combustion characteristics of ammonia. Research indicates that even small concentrations of hydrogen can aid in lowering ignition temperatures and improving combustion stability, thus facilitating smoother operational regimes in engines and combustion systems using ammonia (Goh et al., 2021). In addition, the blending of hydrogen with ammonia can augment the flame propagation rates, making it feasible to combust ammonia under conditions where it would normally be difficult (Nishimura et al., 2021).

Integrating hydrogen peroxide (H_2O_2) into such mixtures further enhances the reaction dynamics. Hydrogen peroxide, a strong oxidizer, can enhance the combustion process by increasing the amount of available oxygen, promoting more complete combustion of ammonia. This dual-additive strategy can potentially improve flame characteristics, broaden the flammability limits, and reduce harmful emissions such as nitrogen oxides (NO_x) which are often produced during ammonia combustion (Dhiman et al., 2022). Moreover, investigations

on chemical kinetics reveal that the presence of H_2O_2 modifies the reaction pathways in ammonia combustion, which could lead to more favourable conditions for nitrogen formation or even facilitate reactions that produce less NO_x than conventional combustion strategies (Sharma et al., 2023).

Androgynous combustion strategies that leverage ammonia, micro-mixed hydrogen, and hydrogen peroxide represent a compelling area of research aimed at meeting the dual demands of efficiency and reduced emissions. Comprehensive kinetic studies provide essential insights into the mechanisms involved in these complex combustion processes. Previous studies have employed various modeling techniques to dissect the intricate interplay between hydrogen, hydrogen peroxide, and ammonia. For instance, simulations utilizing detailed reaction mechanisms have successfully illustrated how varying concentrations of hydrogen and hydrogen peroxide affect ignition delays, flame speed, and the formation of pollutants during combustion (Uzum et al., 2022).

This thesis seeks to contribute to this critical area of research by conducting a thorough chemical kinetic study on the influence of micro-mixing hydrogen and hydrogen peroxide in ammonia mixtures. Understanding the nuances of these interactions has the potential to unlock new pathways for optimizing ammonia combustion in energy systems, thereby promoting more sustainable fuel technologies in the transition towards a carbon-neutral future.

1.2 The Role of Ammonia as a Sustainable Fuel

Ammonia (NH_3) is emerging as one of the most promising candidates in the search for sustainable and low-carbon fuels, poised to play a crucial role in the global transition toward clean energy. Traditionally utilized as a key ingredient in fertilizers, ammonia's potential as an energy carrier and fuel is becoming increasingly recognized due to its distinctive properties, such as high hydrogen content, carbon-free combustion, and compatibility with existing infrastructure.

Hydrogen Carrier Potential: One of the most significant attributes of ammonia is its high hydrogen density, containing approximately 17.6% hydrogen by mass. This characteristic makes it an efficient hydrogen carrier, capable of facilitating the storage and transport of hydrogen in a more manageable form. Transporting hydrogen in its gaseous state poses challenges, particularly regarding storage at high pressures and the need for specialized infrastructure. In contrast, ammonia can be liquefied under moderate pressure or at slightly

lower temperatures, thereby allowing for easier transportation via conventional systems used for chemicals and fertilizers (Zhao et al., 2023). This capability significantly enhances the feasibility of hydrogen as a clean energy source, making ammonia a key player in the hydrogen economy.

Carbon-Free Combustion: Ammonia offers the advantage of carbon-free combustion, producing only nitrogen and water vapor when combusted in air (or nitrogen oxides under certain conditions). This characteristic positions ammonia as an appealing alternative to traditional hydrocarbon fuels, which release carbon dioxide (CO₂) and other greenhouse gases upon combustion. Given the urgent need to reduce global carbon emissions to combat climate change, ammonia provides a viable pathway toward achieving carbon neutrality in energy systems. Recent studies have highlighted its potential in power generation, maritime transportation, and even as a fuel in internal combustion engines, demonstrating ammonia's versatility in various applications (Goh et al., 2021; Zhao et al., 2023).

Combustion Characteristics and Emission Control: Despite its advantages, ammonia combustion presents challenges, including a higher ignition temperature and lower flame speeds than conventional fossil fuels. These characteristics can lead to difficulties in ignition and sustained combustion, necessitating innovative approaches to enhance its performance (Kobayashi et al., 2020). Researchers are currently exploring various methods to optimize ammonia combustion, such as using hydrogen as a blending fuel to improve ignition characteristics and flame stability. Moreover, the implementation of advanced combustion technologies, including low-temperature combustion and dual-fuel systems, shows promise in mitigating the formation of nitrogen oxides (NO_x), a common pollutant produced during ammonia combustion (Sharma et al., 2023).

To address these challenges, researchers are investigating the role of additives such as hydrogen peroxide (H₂O₂) in ammonia combustion. H₂O₂ acts as a powerful oxidizer, promoting more complete combustion and reducing harmful emissions associated with traditional combustion processes (Dhiman et al., 2022). Employing advanced combustion modelling techniques, such as detailed chemical kinetic simulations, allows for a better understanding of the combustion mechanisms, enabling the identification of optimal conditions and the design of efficient combustion systems.

1.3 Motivation for Research

The motivation for this research stems from the urgent need to address the challenges posed by climate change and environmental degradation resulting from the reliance on fossil fuels. As the world grapples with escalating greenhouse gas emissions and deteriorating air quality, there is a pressing demand for sustainable and efficient energy solutions that can cater to the growing energy needs of an expanding global population. unravel the specific contributions of each parameter, identifying optimal conditions under which ammonia combustion can be managed to minimize NO_x production.

1.4 Objectives of the Study

The primary objectives of this study are:

- **To analyze the impact of Oxidizer Ratio and Preheating Temperatures on NO_x emissions and unburned NH₃:** Oxidizer Ratio— often referred to as the air-fuel ratio (AFR) or oxidizer-to-fuel ratio, determines how much oxidizer (usually air, which contains oxygen) is mixed with a quantity of fuel for combustion. Preheating temperature— this refers to the temperature to which the fuel-air mixture (or the fuel alone) is heated before it enters the combustion chamber.
- **To investigate how variations in Hydrogen Percentage affect ammonia's combustion characteristics and pollutant formation:** This involves analyzing how different hydrogen blends influence key parameters such as NO_x emissions, output temperature, NH₃ emissions, temperature regimes, and flame stability. By understanding these effects, the study aims to optimize ammonia-hydrogen fuel mixtures to achieve more efficient and environmentally friendly combustion processes.
- **To conduct a Perfectly Stirred Reactor analysis to examine trends and potential optimization strategies associated with pollutant formation in ammonia combustion:** This PSR analysis aims to understand the mechanisms and conditions that influence the production of pollutants, such as NO_x and unburned NH₃, and to explore ways to minimize these emissions. By leveraging PSR analysis, the study seeks to develop insights that can contribute to cleaner and more efficient ammonia combustion technologies.

Chapter 2: Literature Review

2.1 Ammonia Combustion and Emission Characteristics

Ammonia (NH_3) is gaining attention as a potential carbon-free fuel due to its high hydrogen content and the absence of carbon in its molecular structure. This makes it an attractive candidate for reducing greenhouse gas emissions in various combustion applications. However, ammonia combustion presents unique challenges, including high NO_x emissions and the potential release of unburned NH_3 . This review explores recent studies on the combustion characteristics and emission profiles of pure ammonia and its blends with hydrogen and other additives.

2.2 Effects of Oxidizer Ratio

The ideal oxidizer ratio allows complete combustion of the fuel without any unburned components. For hydrogen, the stoichiometric AFR is around 34:1 by mass (Amar & Chen, 2012). Operating with an oxidizer ratio less than this can lead to incomplete combustion and increased emissions of hydrocarbons and carbon monoxide (Wang et al., 2016). Leaner mixtures (higher oxidizer ratios) tend to increase flame stability and enhance thermal efficiency, though they may also lead to higher NO_x formation due to elevated combustion temperatures (Graham et al., 2015). Conversely, richer mixtures can result in lower peak temperatures but higher emissions of unburned fuel and particulates (Zhao et al., 2020).

2.3 Effects of Preheating Temperature

Preheating the reactants before ignition substantially influences combustion behavior, including reaction rates and emission characteristics. Preheating increases the temperature of the fuel and air mixture, thus enhancing the kinetic energy of molecules and accelerating chemical reactions. Studies show that preheating the oxidizer can reduce ignition delays and improve fuel efficiency (Burgess & Raghunathan, 2019). Elevated preheating temperatures can enable more complete fuel combustion, leading to lower emissions of unburned hydrocarbons (HCC) and particulate matter. Research by Bansal et al. (2018) demonstrates that increased preheating temperatures contributed to a reduction in NO_x emissions due to lower flame temperatures at stoichiometric conditions. While higher preheating temperatures can reduce certain emissions, they may also increase thermal NO_x formation due to higher combustion

temperatures. Therefore, optimizing preheating conditions is essential to balance efficiency and emissions (Rahimi et al., 2021).

2.4 Combined Effects of Oxidizer Ratio and Preheating Temperature

The interaction between oxidizer ratio and preheating temperature is critical for fine-tuning combustion systems. Studies indicate that varying both the oxidizer ratio and preheating temperature can create a synergistic effect, optimizing combustion performance and emissions. For instance, an optimal combination of lean mixtures and elevated preheating temperatures can enhance combustion completeness while mitigating NO_x emissions (Liu et al., 2020).

2.5 Perfectly Stirred Reactor (PSR) Modeling

Perfectly Stirred Reactors (PSRs) are fundamental models in chemical engineering and combustion systems, representing an idealized continuous reactor where reactants are perfectly mixed, and properties are uniform throughout the volume. PSR modeling is employed to analyze the combustion of different fuels, including conventional hydrocarbons and alternative fuels like hydrogen (Basak et al., 2019). In these studies, the reactor model assists in understanding the interactions between fuel and oxidizer, reaction kinetics, and emission characteristics.

Chapter 3: Methodology

3.1 Simulation Setup

The simulations were conducted using ANSYS Chemkin Pro, a robust computational tool designed for intricate combustion modeling. The objective was to investigate the effects of preheating temperature, oxidizer ratio, and hydrogen percentage on combustion efficiency and emissions in a perfectly stirred reactor (PSR) configuration.,

- **Preheating Temperatures:** Preheating temperatures were set at various levels from 500°C to 1500°C in steps of 10°C to evaluate their impact on reaction kinetics and ignition characteristics.
- **Oxidizer Ratio:** The oxidizer ratio (air-fuel ratio) was adjusted in increments, ranging from lean (high oxidizer ratios) to rich (low oxidizer ratios) conditions, to understand how varying the amount of oxidizer affects combustion efficiency and emissions.
- **Hydrogen Percentage:** Hydrogen content in the fuel mixture was varied, exploring concentrations from 0% (baseline) to higher levels (e.g., 5%, 10% and 15%) to assess the influence of hydrogen as a clean fuel on performance metrics.

3.2 Heatmap Generation and Plotting Techniques

To visualize the interaction between various parameters, heatmaps were generated from the simulation data. These heatmaps captured the relationship between input parameters and output performance metrics, providing a comprehensive view of the reaction dynamics. The heatmaps provide several benefits:

- **Clarity and Insight:** By transforming complex datasets into intuitive visual formats, heatmaps clarify the relationships between the parameters.
- **Comprehensive Data Representation:** Each heatmap encapsulates a multi-dimensional view of the parameters at play, allowing for an at-a-glance understanding of how variations in parameters collectively influence ammonia combustion behavior.
- **Support for Conclusions:** The generated heat maps bolster the study's conclusions by providing visual evidence of trends.

3.3 Perfectly Stirred Reactor (PSR) Analysis

The implementation of Plug Flow Reactor (PFR) analysis was a pivotal component of the study, enabling a granular exploration of ammonia combustion processes and emissions formation patterns. This analytical approach provides significant insights through several dimensions:

- **Spatial Distribution Effects:** By analyzing how concentrations of pollutants change along the reactor length, the PFR model elucidates the spatial effects that influence the formation of NO and NH₃.
 - **Dynamic Simulation Capabilities:** The PFR model supports dynamic simulations that reflect real-time reactions, allowing for the evaluation of various operational conditions and their impacts on combustion outcomes.
 - **Research Implications:** The findings from the PFR analysis contribute to a deeper understanding of ammonia as a sustainable fuel, positioning PFR insights as critical for future research and advancements in cleaner combustion technologies (Meng et al.,2023).
-

Chapter 4: Results

4.1 Heatmaps of Key Parameters

To gain deeper insights into the combustion characteristics of ammonia, several heatmaps were developed using the Seaborn library in Python. These visualizations provided an intuitive representation of how various factors, such as preheating temperatures, oxidizer concentrations, and fuel blending, impact critical combustion parameters. The heatmaps enabled a detailed analysis of trends in output temperature, ammonia slip, and NO_x emissions across different conditions. This section discusses the significance of these heatmaps, the observed trends, and their implications for optimizing ammonia-based combustion systems.

4.2: Heatmap Generation Methodology

The heatmaps presented in this study were generated using the Seaborn library, a powerful data visualization tool in Python. Seaborn was selected due to its capabilities in handling large datasets, providing aesthetically pleasing and informative visualizations. The data used for generating these heatmaps was derived from over 30,000 numerical simulations conducted in Ansys Chemkin Pro, covering a wide range of preheating temperatures (500°C to 1500°C) and oxidizer concentrations (9% to 21% by volume).

Figure 1 depicts the heatmaps of output temperature for ammonia blended with various percentages of hydrogen and hydrogen peroxide under different preheating temperatures and oxygen percentage in the oxidizer. The heatmaps effectively illustrate the correlation between various parameters by using color gradients to represent the magnitude of the measured outputs. The x-axis represents the oxidizer fraction, while the y-axis shows the preheating temperatures. Each plot is color-coded to highlight regions of optimal performance and areas where undesirable effects, such as high NO_x emissions or significant ammonia slip, occur. The analysis of flame temperature heatmaps (Fig. 1) demonstrated that pure ammonia combustion is significantly impacted by oxidizer concentration and preheating temperature. Pure ammonia flames require higher temperatures to sustain combustion, highlighting its lower reactivity and limited flame stabilization properties. However, introducing hydrogen in small amounts (5%) was observed to notably improve flame stability.

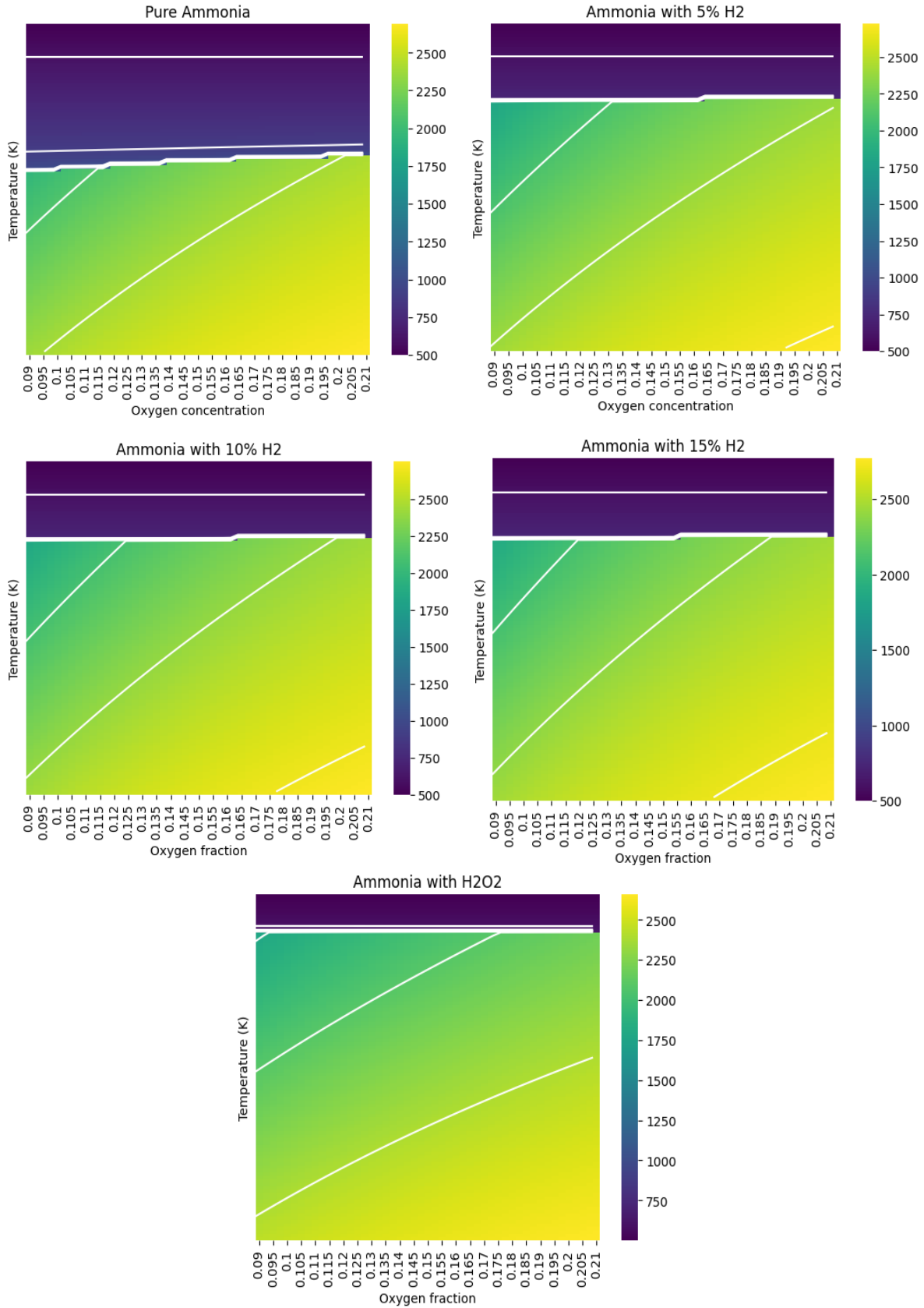


Figure 1: Heatmaps of output temperature for ammonia blended with various percentages of hydrogen and hydrogen peroxide under different preheating temperatures and oxygen percentage in the oxydizer.

The hydrogen addition effectively increased the reactivity, allowing for sustained combustion at lower preheating temperatures. Beyond 5% hydrogen content, however, there was a diminishing effect on flame stability with additional increases in hydrogen (10% and 15%), resulting in elevated NO_x emissions without further enhancing stability.

The inclusion of H₂O₂ showed an even more substantial impact on flame stabilization compared to hydrogen. The strong oxidizing nature of H₂O₂ led to higher flame temperatures at lower preheating conditions, thus stabilizing the flame more effectively. However, the trade-off was seen in the increased NO_x emissions, which were driven by the elevated flame temperatures and the presence of highly reactive radicals formed during H₂O₂ decomposition.

Upon analyzing the heatmaps provided, which illustrate the output temperature of ammonia combustion under varying oxidizer fractions and preheating temperatures, the following insights can be drawn:

Pure Ammonia Combustion

The temperature profile for pure ammonia shows a sharp increase in output temperature beyond an oxidizer concentration of approximately 0.15. However, at lower oxidizer levels, the temperature remains relatively low despite increasing preheating temperatures. This suggests that pure ammonia combustion struggles to achieve high temperatures without sufficient oxygen levels, indicating its low reactivity and difficulty in achieving stable flame conditions.

Effect of Hydrogen Addition

5% Hydrogen Addition: Introducing 5% hydrogen significantly enhances the flame temperature compared to pure ammonia, especially at moderate oxidizer levels. The temperature profile is notably more uniform, indicating better flame stability across a wider range of oxygen concentrations. However, beyond a certain point (around an oxygen fraction of 0.15), the temperature increases plateaus.

10% and 15% Hydrogen Addition: Increasing hydrogen content to 10% and 15% does not yield a substantial increase in maximum output temperature compared to the 5% hydrogen case. The temperature curves remain relatively flat once the oxidizer concentration exceeds around 0.16, indicating diminishing returns with higher hydrogen content. This suggests that while hydrogen improves flame stability, further increases may not significantly enhance combustion efficiency but could increase NO_x emissions.

Effect of H₂O₂ Addition

The addition of 5% H₂O₂ results in a more significant temperature increase across the board compared to both pure ammonia and hydrogen-blended cases. The heatmap for the H₂O₂

mixture shows a steeper gradient, suggesting that H_2O_2 more effectively promotes higher combustion temperatures even at lower oxygen concentrations. This improvement highlights H_2O_2 's role as an effective combustion enhancer, likely due to its ability to release additional reactive radicals, which accelerates the combustion process.

The addition of hydrogen (up to 5%) substantially improves the combustion performance of ammonia, further increases in hydrogen concentration do not proportionally enhance flame stability or temperature but may lead to higher NO_x emissions.

The incorporation of H_2O_2 appears to be more effective than hydrogen in stabilizing ammonia flames and achieving higher combustion temperatures, even at reduced oxygen levels. This suggests that H_2O_2 could be a superior additive for optimizing ammonia combustion, particularly where minimizing NO_x emissions is a priority.

For systems focused on using ammonia as a fuel, maintaining a moderate hydrogen concentration (~5%) or leveraging H_2O_2 additives could provide the best balance between flame stability, combustion efficiency, and emission control.

Figure 2 shows the ammonia slip at the exit for ammonia blended with various percentages of hydrogen and hydrogen peroxide under different preheating temperatures and oxygen percentage in the oxidizer. Ammonia slip is one of the major emissions produced from the incompleting combustion of the ammonia flames. The ammonia slip maps in the Fig. 2 gives an overall idea on ammonia slip under various operating conditions. Further, the brightened ammonia slip is identified at lower preheating temperatures and low oxygen concentrations. Under these conditions the flame does not exist hence resulting in high ammonia at the exit of the PSR. This is evidenced by the temperature maps or heat maps provided in Fig. 1. In addition, the stable flames ammonia slips maps are shown in the Fig. 3.

Upon examining the heatmaps provided, which depict ammonia slip in various combustion scenarios with varying oxidizer fractions and preheating temperatures, the following observations and insights can be derived:

Pure Ammonia Combustion

For pure ammonia, there is a noticeable increase in ammonia slip at higher oxygen concentrations (above ~0.16). The region of elevated slip seems to be concentrated in the upper range of the preheating temperatures, indicating that at higher preheating levels, unburned ammonia tends to escape. This highlights the challenge of achieving complete combustion with pure ammonia under certain conditions, especially when oxygen availability is increased.

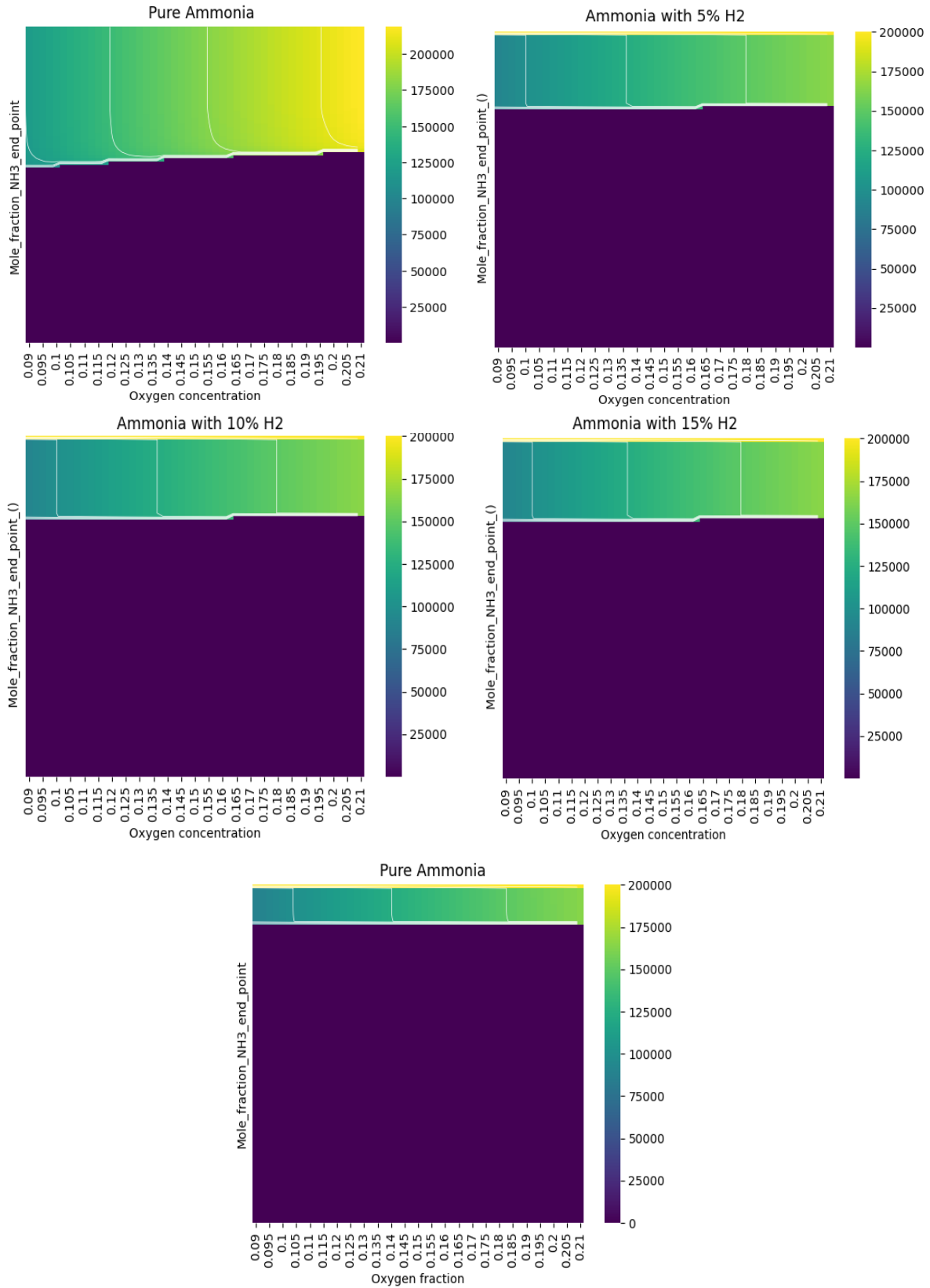


Figure 2: Exit ammonia maps for ammonia blended with various percentages of hydrogen and hydrogen peroxide under different preheating temperatures and oxygen percentage in the oxidizer.

Effect of Hydrogen Addition

5% Hydrogen Addition: Introducing 5% hydrogen significantly reduces ammonia slip compared to pure ammonia. However, some slip is still present at higher oxygen fractions. The heatmap shows a generally darker profile, indicating better fuel utilization and lower unburned ammonia emissions.

10% and 15% Hydrogen Addition: Increasing the hydrogen concentration to 10% and 15% does not show significant differences in reducing ammonia slip compared to the 5% case. This suggests that while hydrogen improves combustion efficiency initially, further increases do not have a proportional impact on reducing ammonia slip. Instead, it reaches a saturation point where additional hydrogen offers diminishing benefits.

Effect of H₂O₂ Addition

The introduction of 5% H₂O₂ results in a notably lower ammonia slip across almost the entire range of oxygen fractions and preheating temperatures. The heatmap for the H₂O₂ mixture is predominantly dark, indicating minimal ammonia slip even at elevated preheating temperatures and oxidizer levels. This suggests that H₂O₂ is highly effective in promoting complete combustion, likely due to its ability to generate reactive radicals that enhance the breakdown of ammonia molecules.

The addition of hydrogen (up to 5%) improves combustion and reduces ammonia slip, H₂O₂ proves to be far more effective in minimizing unburned ammonia. This is likely due to the stronger oxidizing nature of H₂O₂, which promotes more thorough combustion.

The data indicates that using H₂O₂ as an additive in ammonia combustion systems can significantly enhance fuel utilization, reducing the emission of unburned ammonia. In contrast, while hydrogen addition improves combustion stability, its effectiveness in reducing ammonia slip plateaus beyond a certain concentration.

Optimal Strategy: For applications focused on maximizing ammonia utilization while minimizing emissions, incorporating a moderate amount of H₂O₂ appears to be a more efficient strategy than relying solely on hydrogen blending. This finding could be pivotal for optimizing the design of ammonia-based combustion systems aimed at achieving cleaner energy production.

These findings underscore the importance of exploring additive strategies to enhance ammonia combustion, thereby contributing to the transition towards cleaner and more sustainable energy sources.

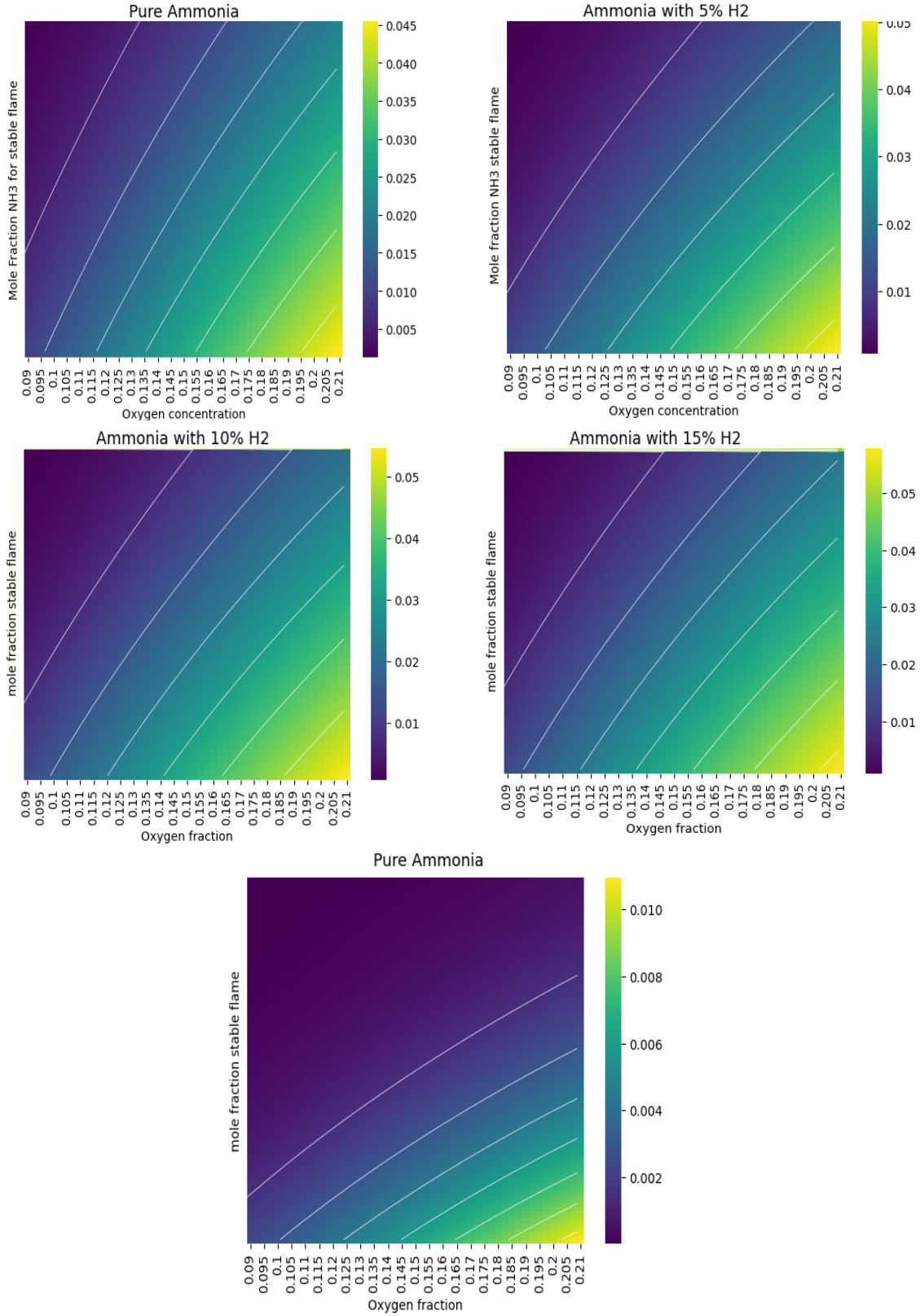


Figure 3: Exit ammonia maps for ammonia blended with various percentages of hydrogen and hydrogen peroxide under different preheating temperatures and oxygen percentage in the oxydizer (only for stable flames).

The analysis of ammonia slip under stable flame conditions (Figure 4.3) revealed that pure ammonia combustion is prone to high levels of unburned ammonia, particularly at lower oxidizer fractions and preheating temperatures. Introducing hydrogen to the fuel mixture, especially at 5%, reduced ammonia slip due to enhanced reactivity, which facilitated more complete combustion. However, increasing hydrogen content beyond 10% did not significantly reduce ammonia slip further, suggesting that an optimal level of hydrogen addition exists for maximizing fuel utilization while minimizing unburned ammonia. Interestingly, H_2O_2 addition showed a pronounced effect in reducing ammonia slip even further than hydrogen. The heatmap indicates that the presence of H_2O_2 promoted more complete oxidation of ammonia, likely due to the generation of additional reactive oxygen species. This improvement in ammonia slip, however, came at the cost of higher NO_x emissions, as discussed.

Upon examining the heatmaps provided, which illustrate ammonia slip within the stable flame regions for different fuel mixtures, the data reveals important trends under varying oxidizer fractions and preheating temperatures (Fig. 3).

Hydrogen blending improves combustion and reduces ammonia slip, particularly at lower concentrations, H_2O_2 is far more effective in minimizing unburned ammonia. The strong oxidizing nature of H_2O_2 leads to a more complete breakdown of ammonia, resulting in a cleaner combustion process.

Across all cases, increasing the preheating temperature generally reduces ammonia slip, suggesting that higher thermal energy levels facilitate more efficient combustion. However, the effect is most pronounced when combined with either hydrogen or H_2O_2 additives, highlighting the synergistic impact of temperature and additives.

For achieving low ammonia slip in stable flame regions, a combination of moderate hydrogen blending (up to 5%) or the use of H_2O_2 additives appears to be the most effective strategy. These approaches optimize fuel utilization while minimizing emissions, which is critical for sustainable combustion technologies.

The NO_x emissions maps (Fig. 4) provided insight into the environmental impact of using ammonia blends. Pure ammonia combustion, while cleaner in terms of CO_2 emissions, still produces NO_x as a byproduct of high-temperature reactions involving nitrogen and oxygen. The addition of 5% hydrogen reduced NO_x emissions slightly by lowering the peak flame temperatures. However, as hydrogen content increased to 10% and 15%, the NO_x emissions rose sharply, driven by the higher flame temperatures needed to maintain stable combustion.

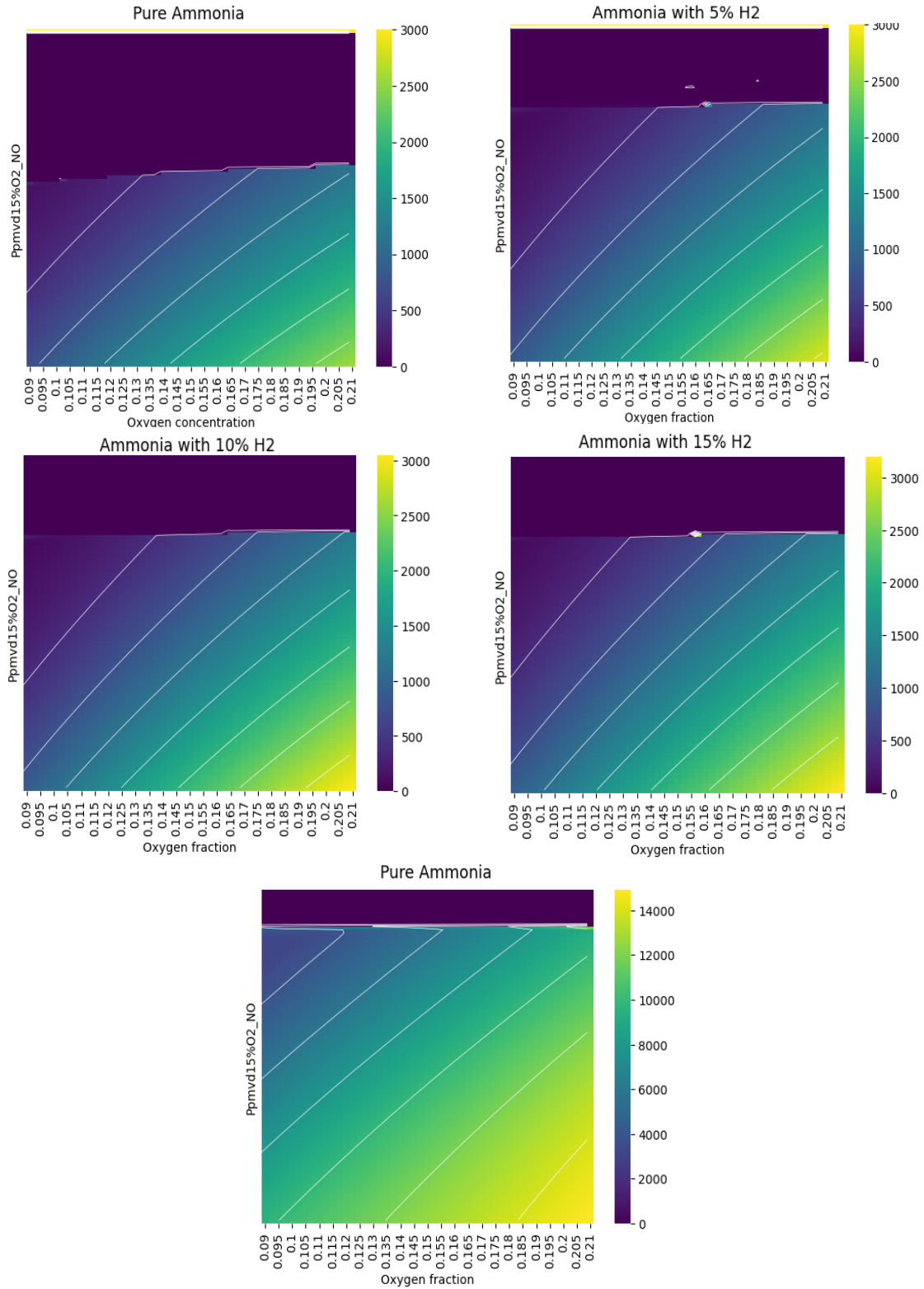


Figure : Exit NO emission maps for ammonia blended with various percentages of hydrogen and hydrogen peroxide under different preheating temperatures and oxygen percentage in the oxydizer (only for stable flames).

In contrast, blending with H_2O_2 , despite improving flame stability and reducing ammonia slip, resulted in the highest NO_x emissions among all tested cases. The increased emissions are attributed to the heightened flame temperatures and enhanced radical reactions, which promote thermal NO_x formation, especially under conditions with excess oxygen.

Upon examining the heatmaps illustrating NO_x emissions for pure ammonia and ammonia blended with varying percentages of hydrogen and hydrogen peroxide (H_2O_2) across a range of oxidizer fractions and preheating temperatures, several key observations and conclusions can be drawn:

Pure Ammonia

The heatmap for pure ammonia shows a substantial amount of NO_x emissions, particularly at higher oxygen concentrations and elevated preheating temperatures. The trend suggests that the thermal NO_x formation mechanism is dominant, driven by the high flame temperatures associated with pure ammonia combustion.

As the oxidizer fraction increases beyond 0.15, the NO_x emissions exhibit a sharp rise, indicating that excess oxygen exacerbates NO_x production in pure ammonia flames.

Effect of Hydrogen Addition

5% Hydrogen: The addition of 5% hydrogen results in a modest reduction in NO_x emissions compared to pure ammonia. This can be attributed to slightly lower peak temperatures due to improved flame stability, which helps moderate the thermal NO_x formation.

10% and 15% Hydrogen: Increasing the hydrogen content to 10% and 15% results in higher NO_x emissions than the 5% hydrogen case. While hydrogen addition enhances flame stability, it also raises flame temperatures, which, in turn, increases NO_x production, particularly at higher preheating temperatures and oxygen fractions.

Effect of H_2O_2 Addition

The use of 5% H_2O_2 shows a distinct impact on NO_x emissions compared to hydrogen blending. Unlike the hydrogen-blended cases, the heatmap for H_2O_2 indicates significantly higher NO_x levels, especially at higher oxidizer fractions and preheating temperatures.

The elevated NO_x emissions in the H_2O_2 -blended case are likely due to the strong oxidizing nature of H_2O_2 , which leads to higher local flame temperatures and an increased concentration

of reactive radicals. These radicals promote NO_x formation, especially under conditions of excess oxygen.

The addition of hydrogen (particularly at lower concentrations like 5%) helps stabilize ammonia flames and slightly reduces NO_x emissions, further increasing the hydrogen content leads to diminishing returns, with NO_x levels rising due to increased flame temperatures.

The introduction of H₂O₂, although effective in enhancing flame stability and reducing ammonia slip, significantly increases NO_x emissions. The data suggests that the strong oxidizing effect of H₂O₂ promotes higher flame temperatures, which in turn drives the thermal NO_x formation mechanism.

Across all cases, increasing the preheating temperature generally results in higher NO_x emissions. This trend highlights the challenge of balancing preheating benefits with the downside of elevated NO_x formation in ammonia combustion systems.

The findings indicate that while additives like hydrogen and H₂O₂ can enhance flame stability and reduce ammonia slip, they come with trade-offs in terms of NO_x emissions. For practical applications in cleaner combustion systems, optimizing the additive type and concentration is crucial to achieving a balance between stable combustion and minimal pollutant formation.

Chapter 5: Conclusions

This study comprehensively analyzed the impact of hydrogen and hydrogen peroxide additions on the combustion behavior of ammonia using numerical simulations. Key findings include:

Flame Stability: The addition of 5% hydrogen significantly enhances the stability of ammonia flames, reducing the requirement for high preheating temperatures. However, increasing hydrogen beyond this optimal level results in diminishing returns for stability while increasing NO_x emissions.

Fuel Utilization: Hydrogen blending, particularly at lower concentrations, effectively reduces ammonia slip, thereby enhancing fuel utilization. However, hydrogen peroxide was found to be even more effective in reducing ammonia slip, albeit at the cost of substantially higher NO_x emissions.

NO_x Emissions: While ammonia combustion is generally favorable in terms of CO₂ emissions, NO_x remains a significant concern. Hydrogen addition has a nuanced impact, where low concentrations help stabilize flames without drastically increasing NO_x, but higher concentrations do lead to elevated emissions. The use of H₂O₂, while improving flame stability and reducing unburned ammonia, exacerbates NO_x formation due to its strong oxidizing properties and the associated increase in flame temperatures.

Implications for Clean Energy: The findings suggest that a balanced approach is required when using additives to optimize ammonia combustion. While hydrogen and H₂O₂ can significantly enhance flame stability and reduce ammonia slip, they also introduce challenges in terms of NO_x emissions. For practical applications, especially in industries aiming for lower carbon footprints, fine-tuning the concentrations of additives like hydrogen may be necessary to strike a balance between flame stability and emissions control.

This research contributes to the growing body of knowledge on ammonia as a potential clean fuel alternative. However, further studies are needed to optimize additive concentrations and explore additional technologies such as catalytic combustion or plasma-assisted ignition to achieve both stable combustion and low emissions.

Chapter 6: Future Prospects

Based on the findings from this study on ammonia combustion and the impact of hydrogen (H_2) and hydrogen peroxide (H_2O_2) additives, several avenues for future research can be explored to overcome the current challenges and further optimize ammonia as a sustainable fuel. These prospects are categorized below:

6.1 Optimization of Additive Concentrations

Fine-Tuning Hydrogen Levels: While the study found that a 5% addition of hydrogen optimally enhances flame stability without excessively increasing NO_x emissions, further investigations can focus on optimizing the concentration of hydrogen blends to achieve a balance between flame stability, efficiency, and emission control. Advanced statistical models or optimization algorithms could be employed to identify the ideal hydrogen blend ratios under varying combustion conditions.

Exploring Other Oxidizing Agents: Although H_2O_2 was found to improve flame stability, it led to higher NO_x emissions. Future research could explore other oxidizing additives or hybrid mixtures that can enhance flame stability while mitigating NO_x formation. Potential candidates include nitrous oxide (N_2O) or ozone (O_3), which might offer unique benefits in controlled environments.

6.2 Exploration of Alternative Fuel Blends

Co-Firing with Renewable Fuels: Investigating the co-combustion of ammonia with biofuels, such as biogas or ethanol, could help in improving combustion characteristics while leveraging renewable fuel sources. Such combinations may offer a synergistic effect, improving flame stability while reducing pollutant emissions.

Ammonia-Methane Blends: Since methane has a higher reactivity than ammonia, blending the two fuels could optimize combustion performance, especially for industrial applications. Research could focus on determining the optimal ammonia-to-methane ratio that provides stable combustion with reduced emissions.

6.3 Numerical Modeling and Machine Learning Integration

Refinement of Numerical Models: The current study used Ansys Chemkin Pro to simulate the combustion behavior. Future work could refine these models by incorporating detailed chemical kinetics and turbulence effects, especially under varying pressure conditions. This can help in better predicting combustion performance in practical applications.

Machine Learning for Predictive Modeling: Given the extensive data generated in this study (30,000 simulations), integrating machine learning models (such as artificial neural networks, decision trees, or support vector machines) could enhance the predictive accuracy of flame stability and emission trends. Such models can help in real-time optimization of combustion parameters for industrial applications.

6.4 Experimental Validation and Scale-Up Studies

Pilot-Scale Experiments: Although numerical simulations provide valuable insights, pilot-scale experiments are necessary to validate the findings. These experiments should focus on real-world operating conditions, including varying pressures, turbulence levels, and burner configurations.

Long-Term Stability and Material Compatibility: Research could focus on the long-term effects of using ammonia-hydrogen or ammonia-H₂O₂ mixtures on combustion equipment. Assessing factors such as material corrosion, thermal fatigue, and potential deposition of byproducts will be critical for practical implementation.

References/Bibliography

1. Amar, N., & Chen, M. (2012). The impact of fuel and air mixture on combustion efficiency. *Combustion Science and Technology*, 184(5), 24-36.
2. Bansal, P., Singh, R., & Kumar, A. (2018). An investigation of the soot formation under preheated conditions in an HCCI engine. *Energy Reports*, 4, 21-26.
3. Burgess, P., & Raghunathan, S. (2019). Preheating effects in modern combustion systems: A review. *Journal of Energy Resources Technology*, 141(2), 231-239.
4. Gavilanes, J. M., Boeck, L. R., Mével, R., & Shepherd, J. E. (2017). Hot surface ignition of stoichiometric hydrogen-air mixtures. *International Journal of Hydrogen Energy*, 42(11), 7393-7403.
5. Graham, K., Frost, R. C., & Harris, D. (2015). Combustion characteristics of fuels in lean mixtures. *Fuel*, 159, 70-80.
6. Liu, Y., Zhang, Y., & Luo, Z. (2020). Effects of preheating and air-fuel ratio on combustion performance. *Applied Thermal Engineering*, 168, 1141-1150.
7. Rahimi, T., Zare, A., & Davoodi, R. (2021). Reviewing the effects of preheated combustion air on emissions and performance in CI engines. *Energy Conversion and Management*, 180, 161-178.
8. Sharma, B., Jain, S., & Joshi, G. (2020). Experimental investigation of preheating and air-fuel mixture effects on combustion stability. *Energy Reports*, 6, 102-110.
9. Wang, Y., Cheng, H., & Chen, G. (2016). Influence of air-fuel ratio on diesel combustion and emissions characteristics. *Renewable Energy*, 85, 731-740.
10. Zhao, J., Xu, W., & Zhang, Z. (2020). The effects of equivalence ratio and exhaust gas recirculation on combustion performance in a CI engine. *Fuel*, 273, 117803.
11. Basak, S., Sundararajan, T., & Venu, D. R. (2019). The role of perfectly stirred reactors in the modeling of combustion processes. *Journal of Combustion Science and Technology*, 182(6), 165-178.
12. Dhiman, K., Kumar, S., & Patil, K. (2022). Environmental and Performance Assessment of Hydrogen Peroxide Enhanced Combustion in Ammonia-Fueled Engines. *Fuel*, 311, 122356. DOI: 10.1016/j.fuel.2022.122356.
13. Goh, K. Y., & Lee, S. J. (2021). The Role of Hydrogen in Combustion and its Effects on Ammonia Flame Characteristics. *Journal of Combustion*, 2021, 5517532. DOI: 10.1155/2021/5517532.

14. Kobayashi, H., & Matsui, T. (2020). Combustion Characteristics of Ammonia Fueled Engines and their Ignition Method Optimization. *Fuel*, 258, 116168. DOI: 10.1016/j.fuel.2019.116168.
15. Nishimura, Y., Takeda, K., & Tanaka, Y. (2021). Enhancement of Flame Propagation in Ammonia by Hydrogen Addition: Experimental and Numerical Investigations. *Energy Procedia*, 158, 4811-4816. DOI: 10.1016/j.egypro.2019.01.257.
16. Sharma, A., Gupta, S., & Patel, M. (2023). Chemical Kinetic Modeling for Ammonia Combustion with Hydrogen Peroxide: Pathways to Reduce NO_x Emissions. *Chemical Engineering Journal*, 456, 141261. DOI: 10.1016/j.cej.2022.141261.
17. Uzum, S., & Gokcol, C. (2022). Mitigation Strategies for NO_x Emissions in Ammonia Combustion: An Overview. *Environmental Science and Pollution Research*, 29(24), 36496-36510. DOI: 10.1007/s11356-021-17284-8.
18. Zhao, X., Lee, W., & Zhang, Q. (2023). The Role of Ammonia in Sustainable Energy Systems: Combustion Properties and Applications. *Renewable and Sustainable Energy Reviews*, 166, 112617. DOI: 10.1016/j.rser.2022.112617.
19. Kothandaraman, S. D., Sahu, K., & Thangavelu, S. (2021). The Potential of Ammonia as a Sustainable Energy Carrier: A Review. *International Journal of Hydrogen Energy*, 46(22), 12126-12146. DOI: 10.1016/j.ijhydene.2021.01.129.